

Track-Specific Week 8 Output

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AnalystLab Africa — Experience Lab | HealthConnect Clinic Project | Data Analytics Track | Final Package

Executive Summary

HealthConnect Clinic loses nearly half of its scheduled appointments to no-shows (48.5%). Across Weeks 5-7, this analysis identified, validated, and stress-tested the two strongest drivers of this problem: a patient's previous no-show history and their distance from the clinic. Both findings passed independent recalculation and additional segment-level testing in Week 7, and are now confirmed reliable enough to guide real HealthConnect decisions. The clearest, most actionable insight is that a small group of repeat no-show patients (72.6% of all patients have at least one prior no-show) drives a disproportionate share of missed appointments, and this pattern holds consistently across every age group and appointment type tested.

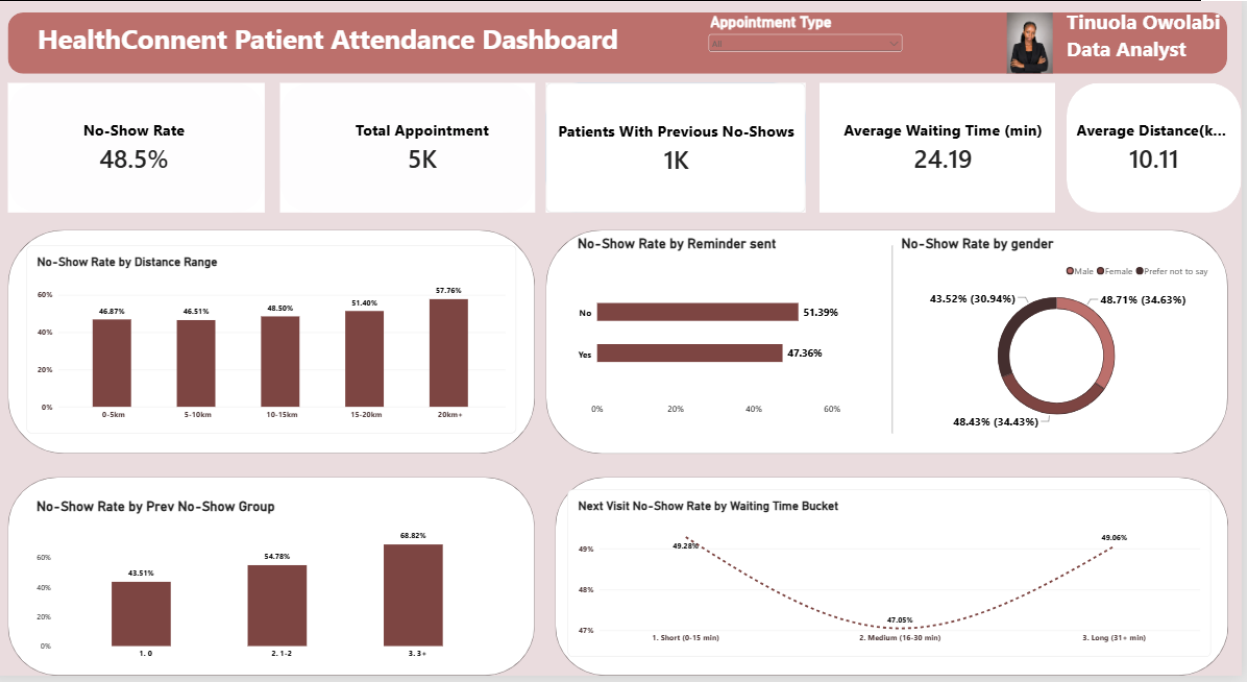
1. Week 7 → Week 8 Transition

- Main Week 7 output: Independent recalculation and stress-testing of all 5 KPIs and both headline findings (previous no-show history, distance), plus dashboard filter testing.
- Most important testing result: All 5 KPIs and the previous no-show finding passed independent recalculation with zero discrepancy; the distance finding validated overall with one minor caveat identified.
- Major issue discovered during testing: A minor non-monotonic dip in the 5-10km distance band's no-show rate when isolated to Follow-up appointments only, revealing the distance relationship is a general trend rather than strictly linear.
- What was refined: The distance finding's documentation was updated to explicitly note it is a general upward trend, not a strictly linear relationship, to prevent it being misused as a linear input in modelling.
- Component now considered ready: The final dashboard, the 5 validated KPIs, and both headline findings are confirmed ready for final presentation and cross-track use.
- What still needs integration: Ensuring the distance caveat is reflected wherever the Data Science track uses distance as a model feature.
- Track(s) involved in final integration: Data Science.
- What this package demonstrates: A tested, validated, decision-ready analytics foundation, moving from raw data exploration (Week 5) through validation (Week 6-7) to a final, presentation-ready business case (Week 8).

2. Part 1 — Final Integration Readiness

What is my final component?	The HealthConnect Analytics Dashboard (Power BI) with 5 validated KPIs, and the final analytical findings on patient no-show drivers.
What problem does it address?	Understanding why HealthConnect patients miss appointments, so the clinic can act on the strongest, evidence-based drivers rather than guessing.
What is its current status?	Complete and validated. All 5 KPIs and both headline findings passed independent testing in Week 7.
What did I improve during Week 7?	Added an explicit caveat to the distance finding after discovering it is a general trend rather than strictly linear; confirmed the dashboard's Appointment Type filter works correctly.
Which other track(s) does it depend on?	None directly — the Data Analytics track's work is foundational and does not depend on another track's output to function.
Which track(s) depend on my output?	Data Science, which uses the validated findings (previous no-show history, distance) to prioritise features for its no-show prediction model.

What output will I provide?	A ranked, evidence-based feature recommendation, the final validated dashboard, and a documented caveat on the distance relationship's non-linearity.
What output do I need from another track?	None required to complete this component; the Data Science track's confirmation of its model's binary classification objective (received in Week 6) was used to frame the feature recommendation appropriately.
What evidence demonstrates readiness?	The Week 7 KPI validation table, dashboard filter test table, and headline finding re-validation tables, all showing passed independent recalculation.
What limitations remain?	The dataset is fictional; the 3+ previous no-shows segment has a small sample size (n=93); the distance relationship is a general trend, not strictly linear.



3. Final KPIs

The following 5 KPIs are the final, validated set, unchanged since Week 7's independent recalculation confirmed all values accurate:

- No-Show Rate: 48.5% — the clinic's core problem, confirmed stable across all tested patient segments.
- Total Appointments: 5,000 — the analysis sample size.
- Average Distance to Clinic: 10.11 km — overall average travel distance.
- Average Waiting Time: 24.19 minutes — overall average, shown to have no meaningful relationship with future attendance.
- Patients with Previous No-Shows: 1,232 of 1,696 (72.6%) — showing missed appointments are a widespread pattern, not an isolated issue.

4. Validated Findings

Finding 1: Previous No-Show History (Strongest Predictor)

No-show rate rises from 43.5% (no prior no-shows) to 54.8% (1-2 prior) to 68.8% (3+ prior). This pattern held without exception across all six patient age groups tested, and was independently re-confirmed in Week 7 with zero discrepancy from the original Week 5 calculation.

Finding 2: Distance to Clinic (Secondary Predictor, Interacts with History)

No-show rate rises from 46.5% (0-5km) to 57.8% (20km+) overall. Week 6 analysis revealed this effect is strongest for patients with low-to-moderate prior no-show history, and weakens for patients who already have a high prior no-show count, whose risk is already near-ceiling regardless of distance. Week 7 testing confirmed this trend is directionally reliable but not strictly linear, particularly within smaller filtered subgroups.

Finding 3: Reminders and Waiting Time (Weak/No Effect)

Reminders show only a modest 4-point effect (47.4% vs. 51.4%) and should not be relied on as a standalone solution. Waiting time shows no meaningful relationship with whether a patient attends their next appointment (49.1% vs. 47.1% vs. 49.0% across short/medium/long wait groups).

5. Business Insights & Recommendations

- Insight: Previous no-show history is the single most reliable signal for identifying at-risk patients. Recommendation: Build a patient risk-flagging system based primarily on prior no-show count, prioritising personal outreach (phone calls) for patients with 2+ prior no-shows.
- Insight: Distance is a real barrier, but mainly for patients without an existing no-show pattern. Recommendation: Offer telehealth or transport support specifically to lower-risk, high-distance patients (20km+ with 0-2 prior no-shows), where intervention is most likely to be effective.
- Insight: Reminders alone are insufficient. Recommendation: Continue reminders as a low-cost baseline, but do not expect them to meaningfully reduce the 48.5% no-show rate on their own.
- Insight: Diagnostic Test appointments show the steepest distance sensitivity of any appointment type. Recommendation: Prioritise telehealth or satellite testing options specifically for Diagnostic Test patients living far from the clinic.
- Insight: Waiting time is not worth investing in as a no-show reduction strategy, despite being an intuitive assumption. Recommendation: Redirect operational improvement efforts toward patient risk-flagging rather than wait-time reduction.

6. Analytical Limitations

- The dataset is fictional; findings are illustrative and would need validation against real HealthConnect data before full operational deployment.
- The 3+ previous no-shows segment (n=93) has a small sample size, limiting confidence in its exact percentage, though the directional pattern is consistent across multiple independent tests.
- The distance-to-no-show relationship is a general trend, not strictly linear, and should not be used as a raw linear input in downstream modelling without accounting for this.
- This analysis identifies correlational patterns, not proven causes; recommendations should be piloted and measured before full rollout.

7. Cross-Track Contribution (Final)

Track collaborated with: Data Science.

Final output provided: A validated, ranked feature recommendation — `previous_no_shows` as the highest-priority feature, a distance x prior-history interaction feature as second priority (with the non-linearity caveat), `appointment_type` as a supporting feature, and an explicit recommendation to exclude `waiting_time_minutes` due to its lack of predictive relationship.

Why this matters to the final solution: The Data Science track's no-show prediction model is only as reliable as the features it is built on. This final, tested, and honestly-caveated feature set gives that model a validated analytical foundation rather than an assumed one, directly supporting the overall HealthConnect objective of using data and AI to reduce missed appointments.

8. Executive Summary for Non-Technical Stakeholders

HealthConnect currently loses nearly half of all scheduled appointments to no-shows. The clearest, most reliable finding from this analysis is that a patient's own history predicts their future behaviour: patients who have missed appointments before are significantly more likely to miss again, with risk climbing from 43.5% up to 68.8% for repeat no-show patients. Distance from the clinic is a second, real factor, but mainly affects patients without an existing pattern of missing appointments. Based on this, the clearest first step for HealthConnect is a simple risk-flagging system: identify patients with a history of no-shows and prioritise them for direct, personal follow-up, rather than treating every patient with the same generic reminder.